MTS UTTRAKHAND

31 Ans IST Variety Rice

Total price = $25 \times 32 = 800$ R.S

IInd Variety Rice

Total price = $15 \times 36 = 540$ RS

Total cost of both varieties = $800 + 540 = \boxed{1340}$

Now price of mixture = Total weight $\times$ 40.20

$= (25+15) \times 40.20$

$= 40 \times 40.20 = \boxed{1608 \text{ R.S}}$

So profit = $1608 - 1340 = 268$ R.S

Profit % = $\frac{\text{Profit}}{\text{C.P}} \times 100 = \frac{268}{1340} \times 100 = 20\%$.

32 Ans Let distance = x k.m

Then $t_1 = \frac{x}{4}$ and $t_2 = \frac{x}{5}$

and their difference = 15 minutes

$\Rightarrow \frac{x}{4} - \frac{x}{5} = 15 \text{ minutes} = \frac{15}{60}$ hours.

$\Rightarrow \frac{5x - 4x}{20} = \frac{1}{4}$

and $25x - 16x = 20$

$\Rightarrow x = \frac{20}{4} = 5$ k.m.

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34 Ans Let CP Of each article be 1 R.S

C.P of x articles = x R.S

and SP of 20 articles = 20

Profit = (20-x)

$P\% = \frac{20-x}{x} \times 100 = 25$

$\Rightarrow 2000 - 100x = 25x$

and $125x = 2000 \Rightarrow x = \frac{2000}{125} = 16$ Ans

35 Distance = 600 m

Time = 5 minutes = 5 x 60 = 300 seconds.

$\text{Speed} = \frac{\text{distance}}{\text{Time}} = \frac{600}{300} = 2 \text{ m/s} = 2 \times \frac{18}{5} \text{ k.m/h}$

$= \frac{36}{5} = 7.2 \text{ k.m/h}.$

36 Ans Let principal = 100 R.S

Then ammount = 100 x 2 = 200 R.S

So SI = 200 - 100 = 100 R.S.

Time = 8 R.S R = ?

$SI = \frac{P \times R \times T}{100}$

$\Rightarrow 100 = \frac{100 \times R \times 8}{100}$

$\Rightarrow R = \frac{100 \times 100}{100 \times 8} = 12.5\ \%.$

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37 Ans Total Runs in 10 Innings = 32×10 = 320

Now after 11th Inning

Average = 32+4 = 36

Total Runs = 36×11 = 396

So, Runs in 11th inning = 396-320

= 76 R.S.

38 Ans Let A's Salary = x.

$\therefore$ B's Salary = $x + 10000$

and C's Salary = $x + 10000 + 15000 = x + 25000$

again given

$$x + (x + 10000) + (x + 25000) = 65000$$

$$\Rightarrow 3x + 35000 = 65000.$$

and $3x = 30000 \Rightarrow x = \frac{30000}{3} = 10000$ ₹.S.

39 Ans $1767 = x \times 57$

$$\Rightarrow x = \frac{1767}{57} = 31 \text{ Ans}$$

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40 =

40% Of 265 + 35% of 180 = 50% Of x.

$\Rightarrow \frac{40}{100} \times 265 + \frac{35}{100} \times 180 = \frac{50}{100} \times x.$

$\Rightarrow \frac{40 \times 265 + 35 \times 180}{100} = \frac{50}{100} x$

cmd $x = \frac{10600 + 6300}{50} = \frac{16900}{50} = 338$

POSTAL A B